

I2C	Sub SYS	Function	Part Number	I2C Spec.		I2C Slave Address / Write / Read (7-bit mode)	
I2C-0	AP	Flash LED driver	AW36413			0x6B	
I2C-1 (I3C)	AP Sensor Hub	Magnetic Sensor	AK09918C	400 Kbps		0x0C	Write:0x18 / Read:0x19
		Ambient Light Sensor	STK33562	400 Kbps		0x46	Write:0x8C / Read:0x8D
		Proximity Sensor					
		SAR Sensor	SX9328	400 Kbps		0x28	
I2C-2 (I3C)	AP	Rear Camera					
I2C-3	AP	LCM BIAS	SM5109C			0x3E	Write:0x7C / Read:0x7D
		Audio Smart PA	AW87339				
I2C-4 (I3C)	AP	Front Camera					
I2C-5	AP	Charger IC	BQ25601	400Kbps		0x6B	Write:0xD6 / Read:0xD7
I2C-6	AP	BUCK LP4X					

Note : I2C Spec. : Standard mode (100 kbps) and Fast mode (400 kbps), Fast mode Plus (1 Mbps) and High-speed mode (3.4 Mbps)

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Title 02_I2C_ID_Overview

Size
C

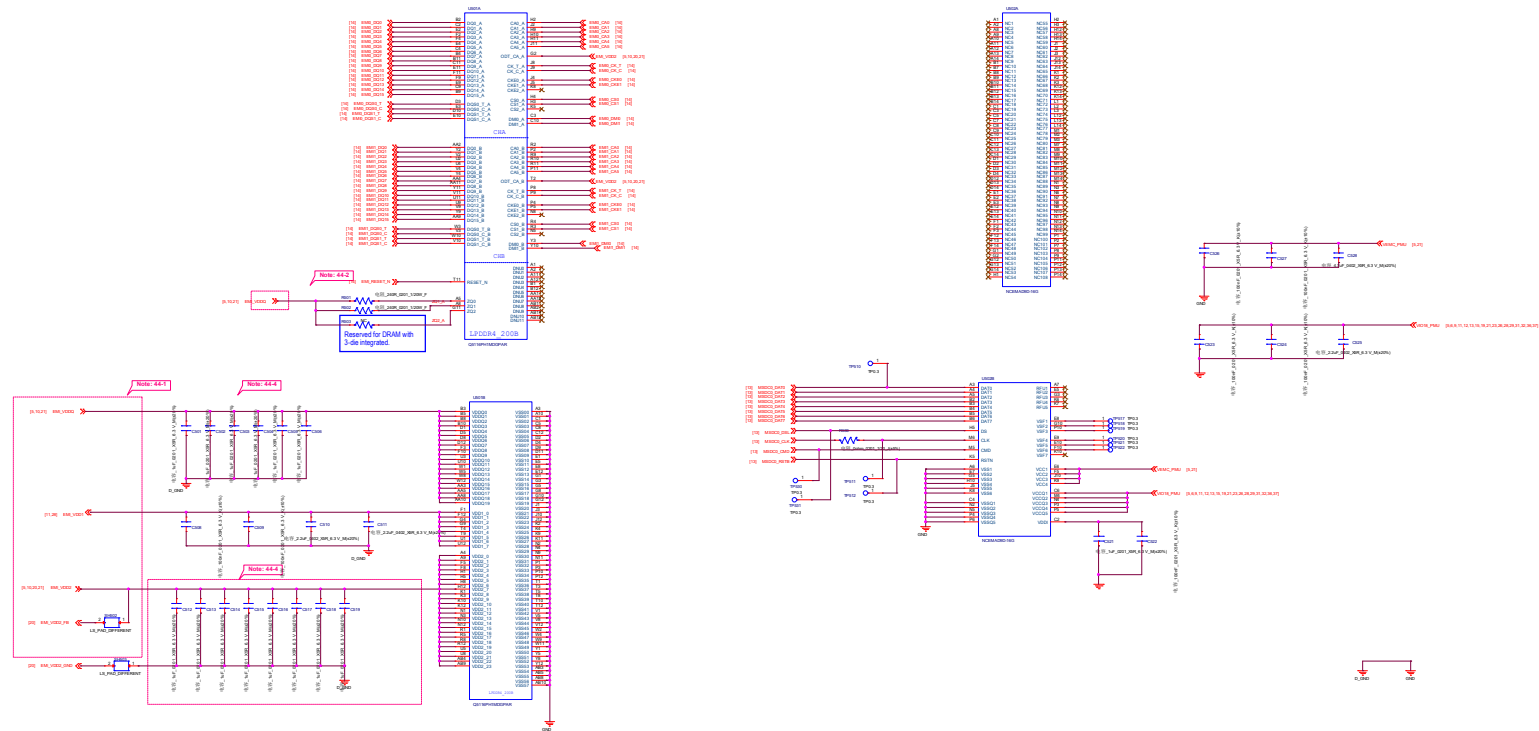
Project PA5905_MB_V1

Rev
V1

Date: Wednesday, July 27, 2022

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华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 04_CHANGELIST		
Size D	Project PA5905_MB_V1	Rev V1
Date: Wednesday, July 27, 2022		Sheet 4 of 37



Schematic design notice of "44_Memory_eMMC_LPDDR4"

Note 44-1: Please refer to power supply related page select output voltage properly for LPDDR4

Note 44-2: DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ.

Note 44-4: VDD2 VDDQ decoupling cap: closed to DRAM ball.
For other cap for PMIC >10uF, at PMIC page;
please also refer to MMD and layout guide for placement.

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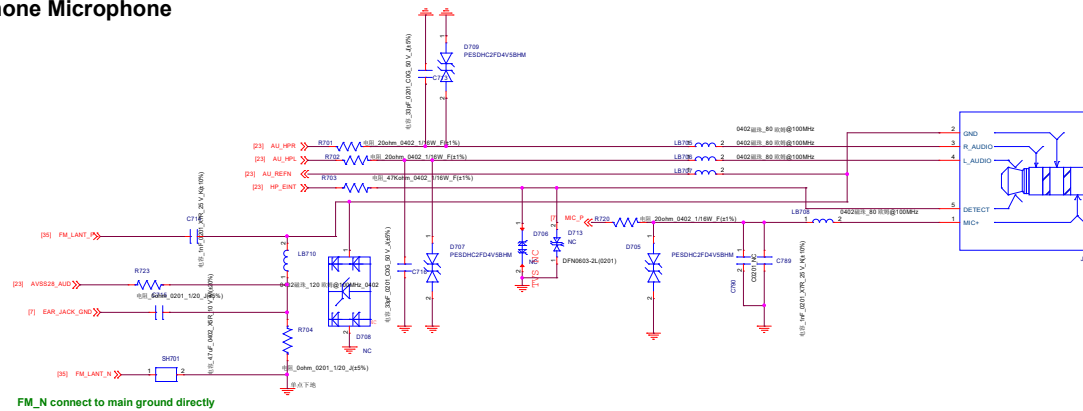
Title 05_Memory_eMMC_LPDDR4

Size E Project PA5905_MB_V1

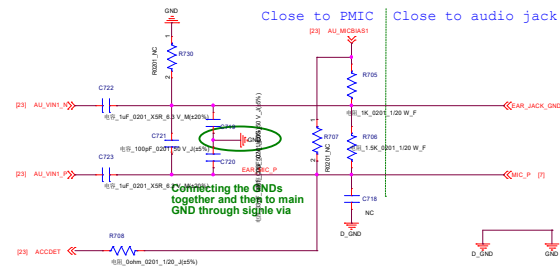
Rev V1

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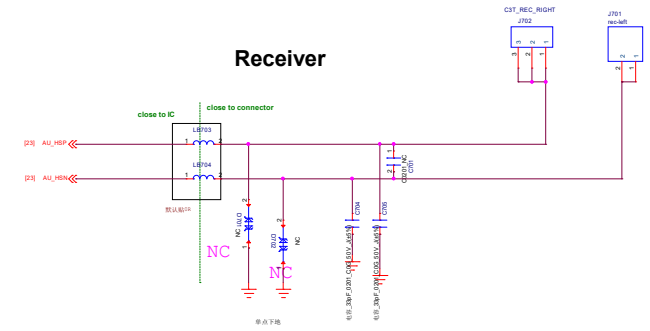
Earphone Microphone



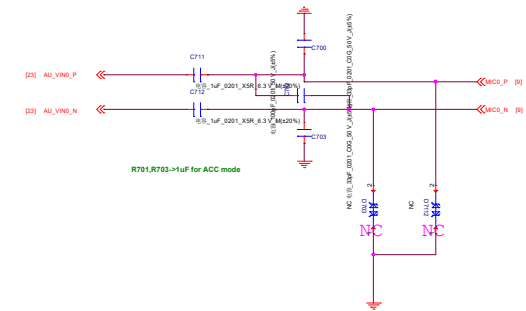
Earphone Microphone



Receiver



Main MIC



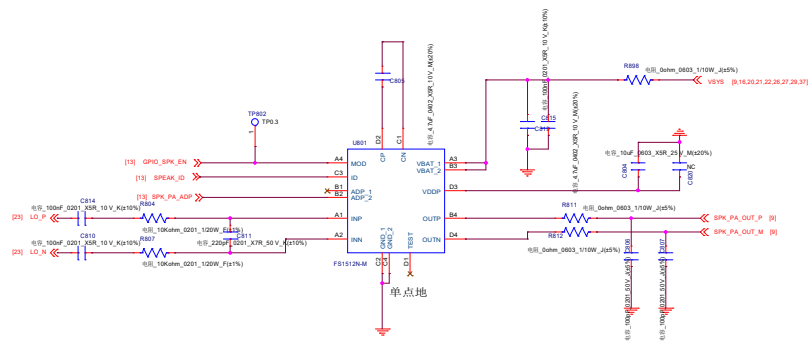
华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 07_AUDIO_REC_MAIN_MIC_HEADPHON

Size D	Project PA5905_MB_V1	Rev V1
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Date: Tuesday, August 30, 2022 Sheet 7 of 37

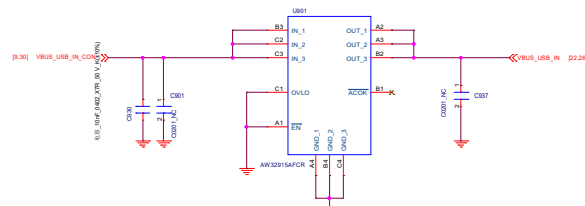
AUDIO PA FS1512N-M



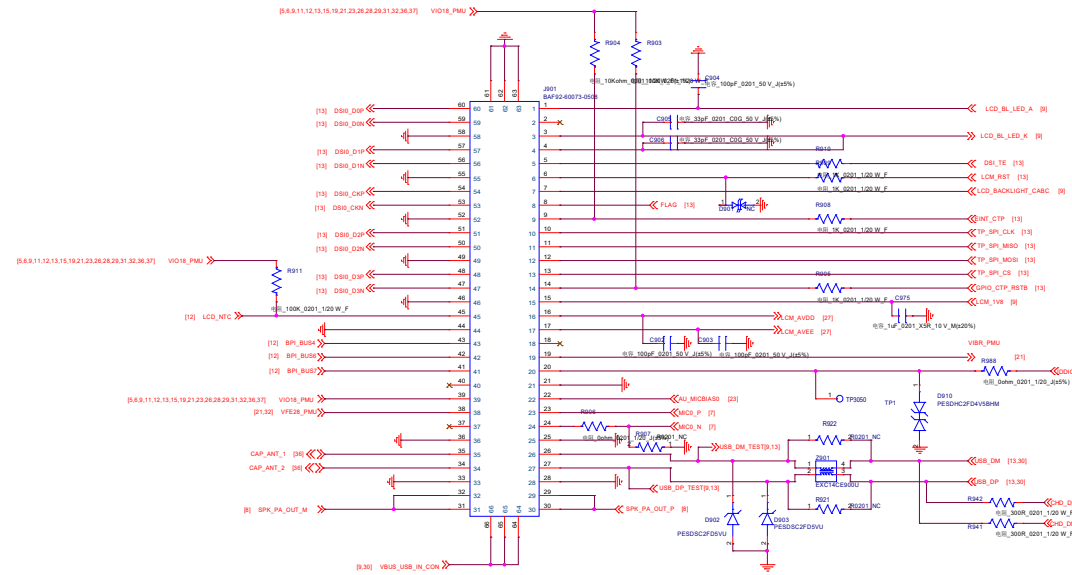
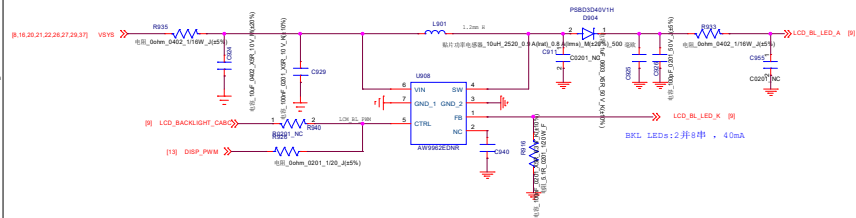
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 08_AUDIO_SPEAKER_AMP			
Size D	Project PA5905_MB_V1		Rev V1
Date: Friday, September 02, 2022		Sheet 8 of 37	

60	Powerpin GND	Powerpin GND
61	Powerpin GND	Powerpin GND
1	LEDA	MIPI_LANE0_P
2	NC	MIPI_LANE0_N
3	LCM_LED_K_1	GND
4	LCM_LED_K_2	MIPI_LANE1_P
5	TP	MIPI_LANE1_N
6	LCM_RST	GND
7	CANC/PMW	MIPI_CLK_P
8	Error Flag PIN	MIPI_CLK_N
9	ENTN CTP	GND
10	SCL	MIPI_LANE2_P
11	MISO	MIPI_LANE2_N
12	MOSI	GND
13	CS	MIPI_LANE3_P
14	GPIO CTP_RSTB	MIPI_LANE3_N
15	LCM_VDDIO	GND
16	LCM_VDDIO	GND
17	LCM_VDDIO	GND
18	NC	PMU_TUNER_RPT_RST0
19	VDDIO PMU	PMU_TUNER_RPT_RST1
20	VDDIO CTP	PMU_TUNER_RPT_RST0
21	GND	ANT_DETECT
22	NC	GPIO
23	TP	TP
24	TP	TP
25	TP	TP
26	TP	TP
27	TP	TP
28	TP	TP
29	TP	TP
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60	TP	TP

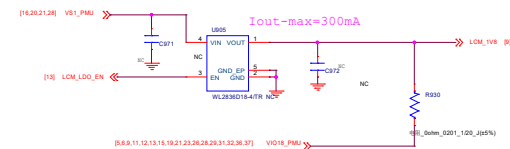
OVP



LCD-BACKLIGHT



屏 单供1.8V NC



华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 09_LCD_CTP_BL

Size D Project

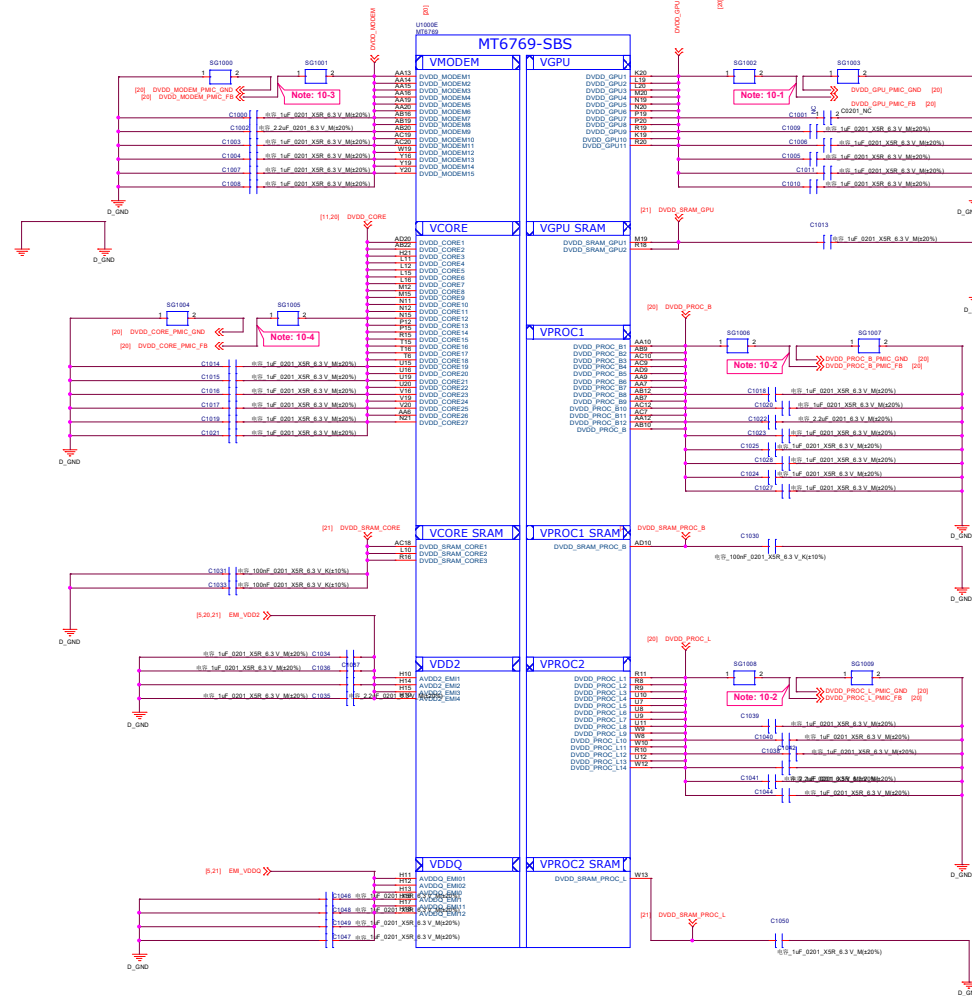
PA5905_MB_V1

Rev V1

Date: Friday, September 02, 2022

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Schematic design notice of "10_BB_POWER_PDN" page:



Note 10-1: Differential pair of DVDD_GPU remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

Note 10-2: Differential pair of DVDD_PROC remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

Note 10-3: Differential pair of DVDD_MODEM remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

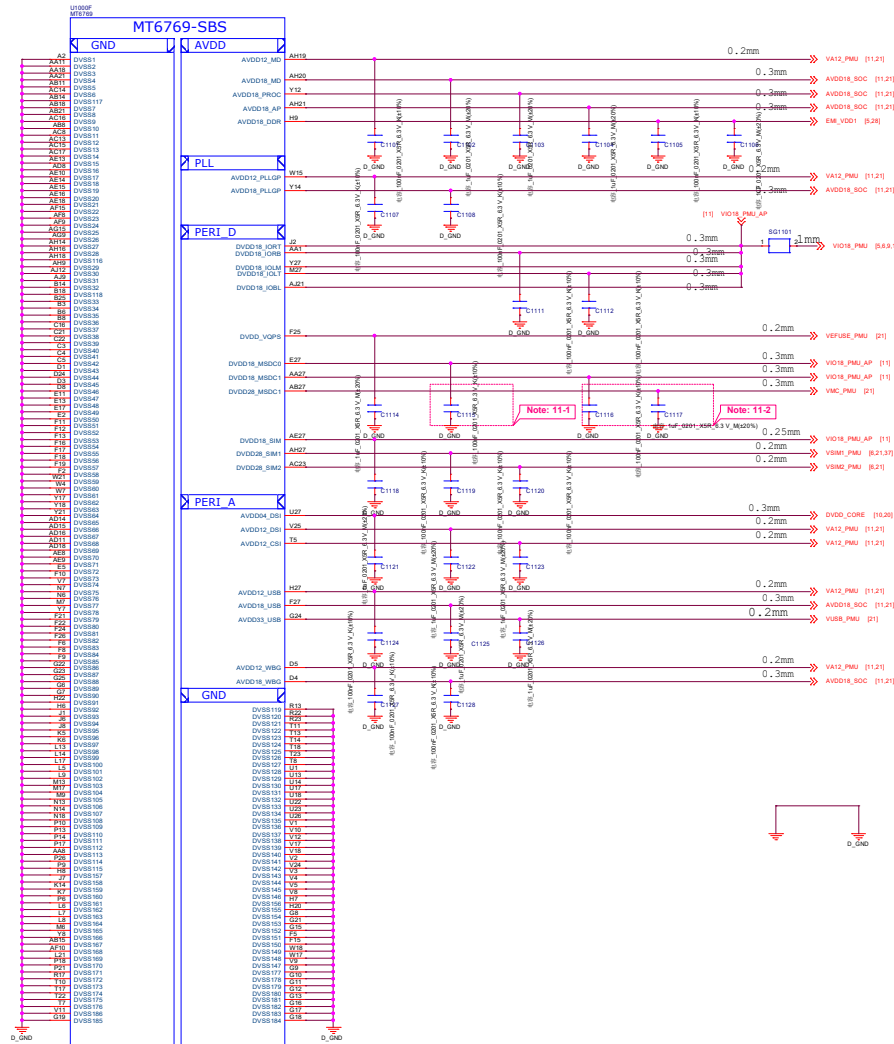
Note 10-4: Differential pair of DVDD_CORE remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

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Title 09_LCD_CTP_BL

Size D	Project PA5905_MB_V1	Rev V1
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Schematic design notice of "11_BB_POWER_IO" page:

Note 11-1: C1115 closed DVDD18_MSDC0 150mil

Note 11-2: C1116 closed DVDD18_MSDC1 150mil
C1117 closed DVDD28_MSDC1 150mil

AVDD12 : 单个点走8mil, 两路分支各16mil, 总路径20mil

VIO18_FMU_AP: 总路径40mil

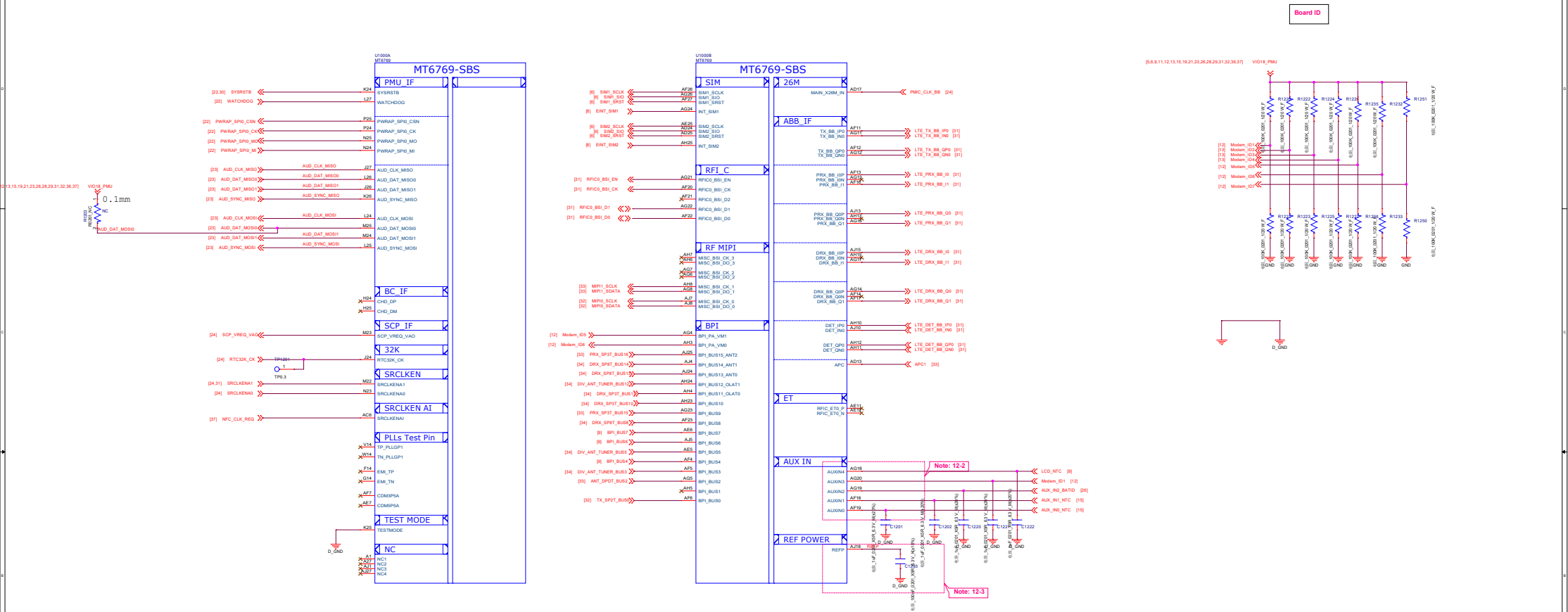
AVDD18_SOC: 总路径20mil

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Title 09_LCD_CTP_BL

Size D	Project PA5905_MB_V1	Rev v1
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Title 09_LCD_CTP_BL

Size	Project	Rev
D	PA5905_MB_V1	V1

Date: Wednesday, July 27, 2022 Sheet 12 of 37

Schematic design notice of "12_BB_1" page:

Note 12-1: PWRAP_SPIO_CSN" and "AUD_DAT_MOSIO" are bootstrap pins to select which interface will be the JTAG pin out

PWRAP_SPIO_CSN	AUD_DAT_MOSIO	AP_JTAG	MD_JTAG
default=PU	default=PD	N/A	N/A
HI	LO	N/A	N/A
HI	HI	SPIO + EINT8	SPI2 + SPI3
LO	LO	SPIO + EINT8	N/A
LO	HI	N/A	N/A

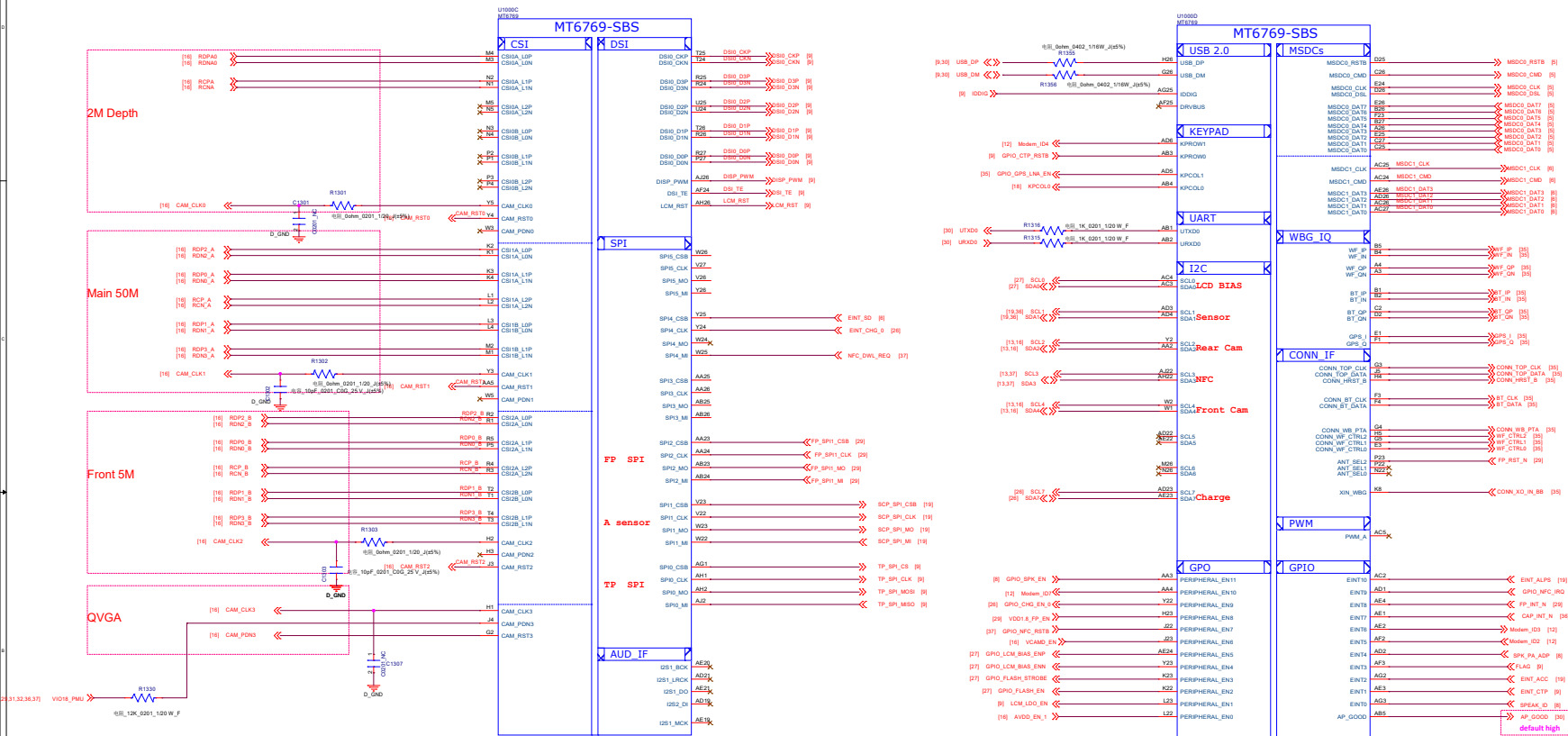
Note 12-2: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

Note 12-3: The de-coupling cap. for REFP (AJ18 ball) have to be placed as close to BB as possible.

Note 12-4: HW pin for DDR type feature in bootstrap (refer to BB HW design Notice)

AUD_SYNC_MISO	AUD_CLK_MISO	CAM_PDN3	PMIC 6358 voltage	DDR Type
default=PD	default=PD	default=PD	VDRAM1 / VDRAM2	DDR
No ext. pull	No ext. pull	No ext. pull	1.125v / 0.6v	LP4X eMCP
No ext. pull	12K pull to VIO18	No ext. pull	OFF / 1.8v	Reserved
12K pull to VIO18	No ext. pull	No ext. pull	1.225v / OFF	LP3 eMCP
12K pull to VIO18	12K pull to VIO18	No ext. pull	1.125v / 1.8v	Reserved

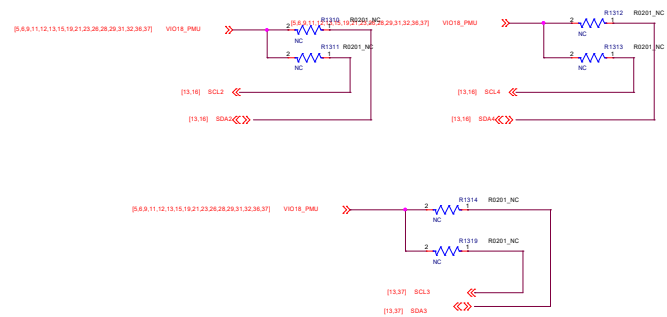
Note 12-5: Charger must have D+/D- pin for charger type USB detection. Charger must have at least 500mA USB current for All charger type.



Design Notice: Boot mode

MT6768/67 DDR trapping table

	MT6768 / MT6767		PMIC MT6358	
	AUD_SYNC_MISO (ball : K26)	AUD_CLK_MISO (ball : J27)	CAM_PDN3 (ball : J4)	VDRAM1
EMCP LP4X (254 b)	No ext. pull	No ext. pull	No ext. pull	1.125V
Reserved	No ext. pull	12K to VIO18	No ext. pull	0V
EMCP LP3 (221 b)	12K to VIO18	No ext. pull	No ext. pull	1.225V
Reserved	12K to VIO18	12K to VIO18	No ext. pull	1.125V
Discrete LP4X (200 b)	No ext. pull	No ext. pull	12K to VIO18	1.125V
Reserved	No ext. pull	12K to VIO18	12K to VIO18	0V
Reserved	12K to VIO18	No ext. pull	12K to VIO18	1.225V
Reserved	12K to VIO18	12K to VIO18	12K to VIO18	1.125V



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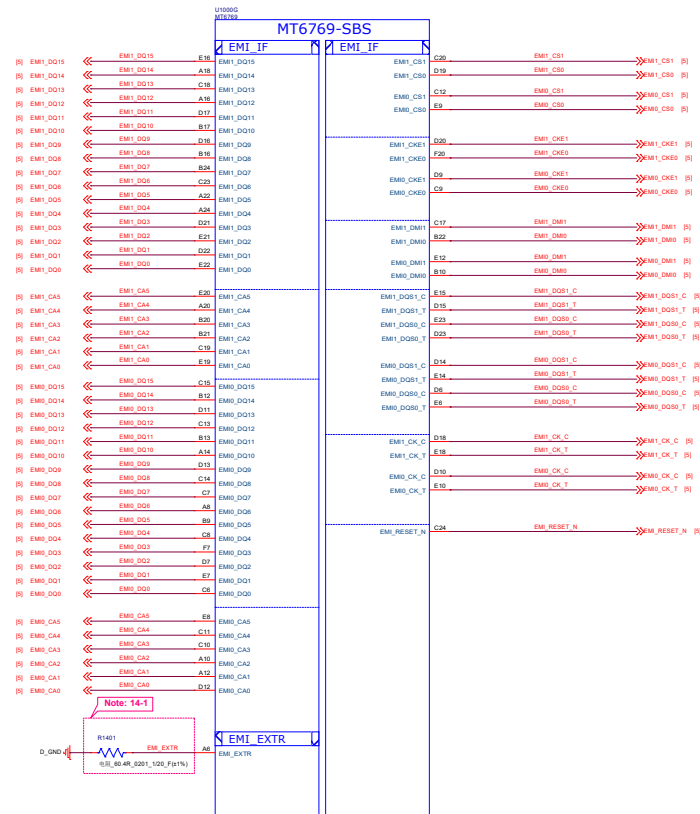
Title 09_LCD_CTP_BL

Size D Project PA5905_MB_V1

Rev v1

Date: Wednesday, July 27, 2022

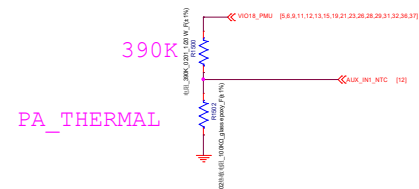
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Schematic design notice of "14_BB_3" page.

Note 14-1: R1401 please select 60.4 ohm (1%) resistor

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 09_LCD_CTP_BL			
Size D	Project PA5905_MB_V1		Rev V1
Date: Wednesday, July 27, 2022	Sheet 14 of 37		

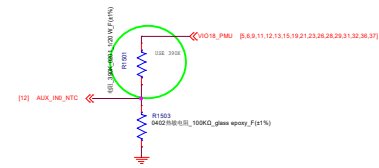


change to NCP03WF104F05RL Murata

Thermistor to sense RF PA temperature

1. NTC1502 must close to LTK Band * PA or the hottest PA <2mm.
2. The distance is the shortest distance from package edge to edge.

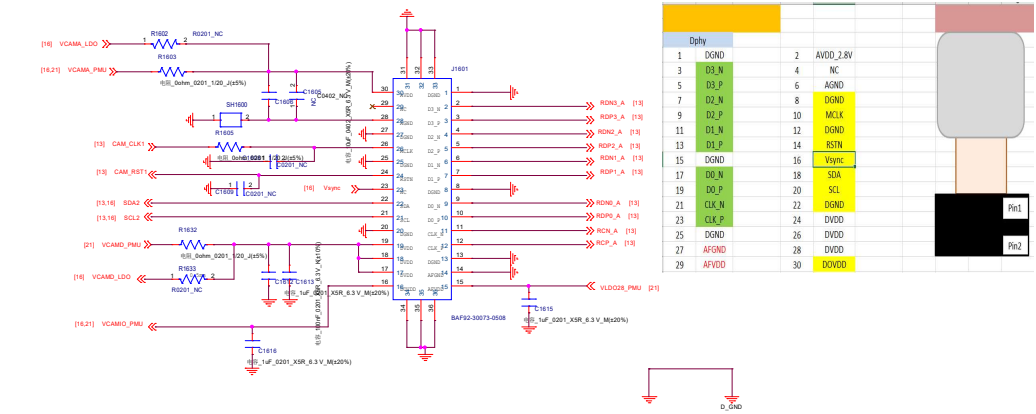
Thermistor / To sense board level temperature



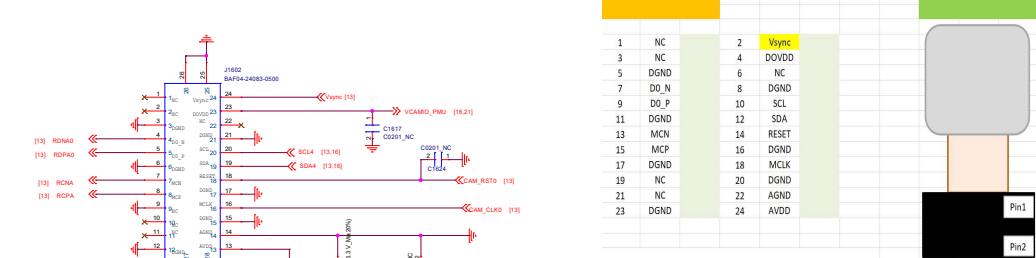
Distance to AP is 5-7mm and away from other heat resource 10mm-12mm

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 15_BB_AUXADC_Thermal		
Size D	Project PA5905_MB_V1	Rev v1
Date: Wednesday, July 27, 2022	Sheet 15 of 37	

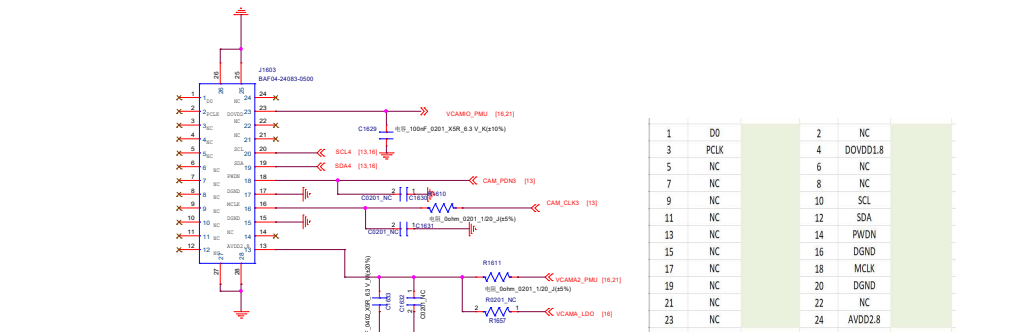
Main Rear Camera - 50M



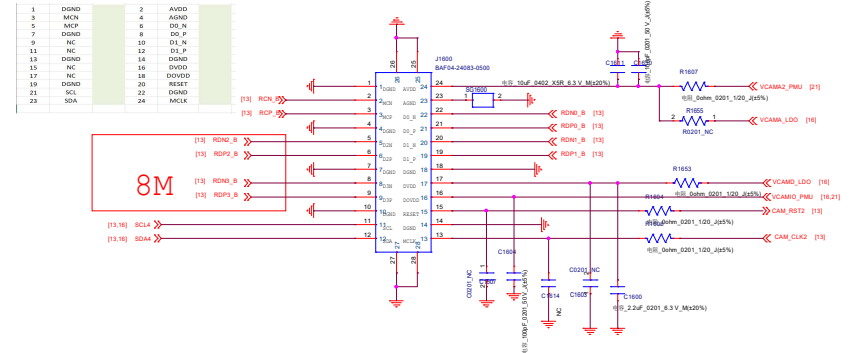
2M (Depth)



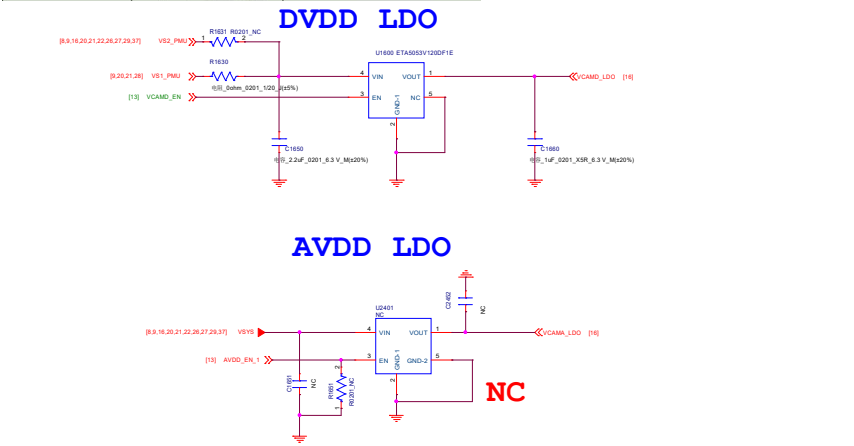
QVGA



Front Camera 5M/8M



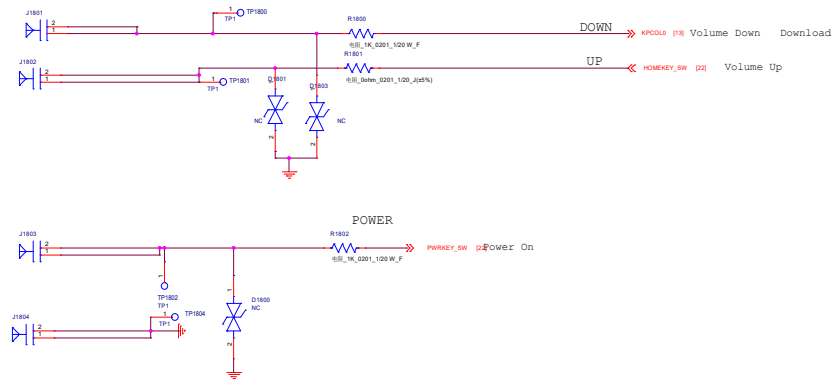
VS1	1.86 ~ 2.2 (12.5mV/step)	2
VS2	1.25 ~ 1.5 (12.5mV/step)	1.35



华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 16_CAMERA_INTERFACE		
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Power Key / Key Pad

DO NOT put pull-up resistor on PWRKEY



Schematic design notice of "65_PERI_Dual_SIM_ICUSB_KEYPAD" page.

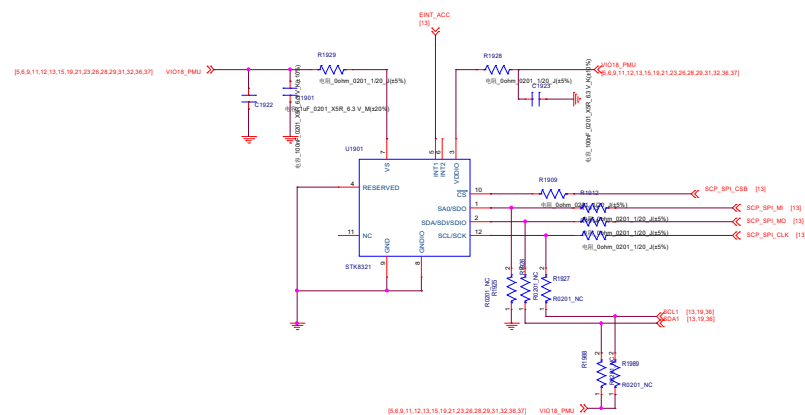
Note 75-1: DO NOT put pull-up resistor on PWRKEY

Note 75-2: Volume Up : HOME Key / GND
Volume Down : KPROW0/KPCOL0

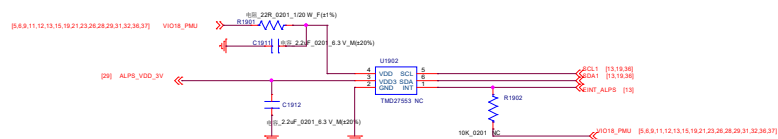
Note 75-3: For better ESD performance, please select suitable components for system protection.

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 18_KEYPAD_INTERFACE		
Size D	Project PA5905_MB_V1	Rev v1
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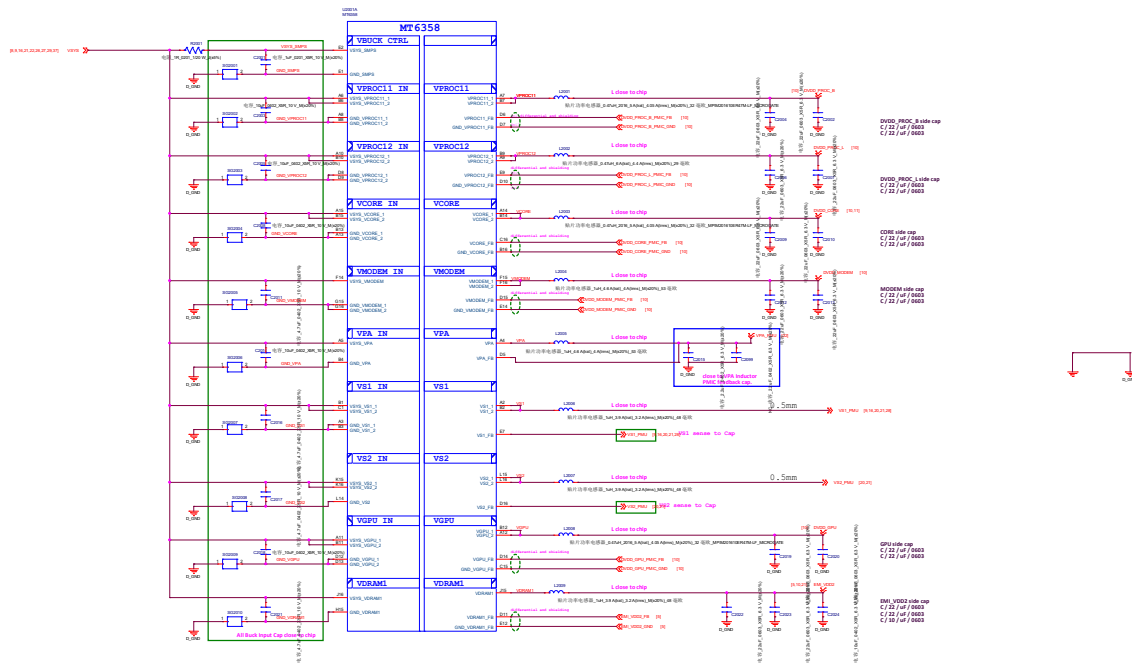
A SENSOR



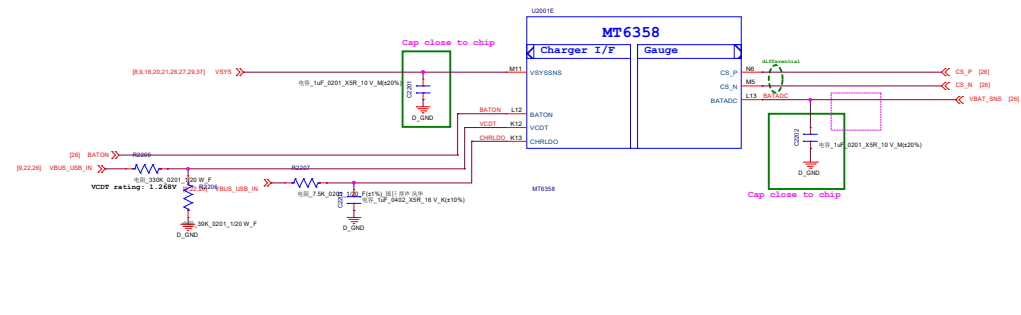
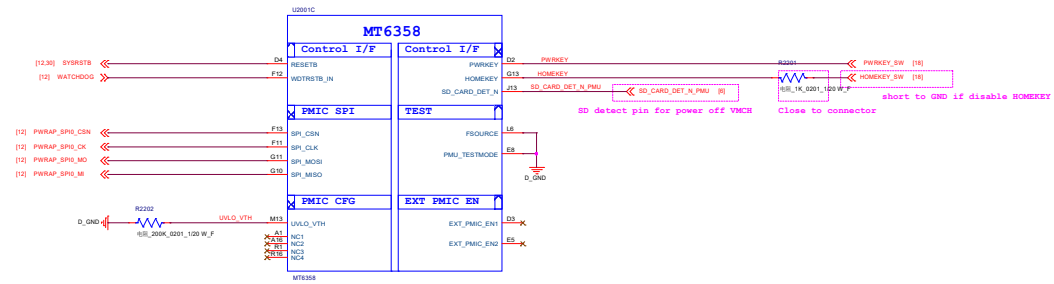
ALS PS 预留 (国内配置贴)



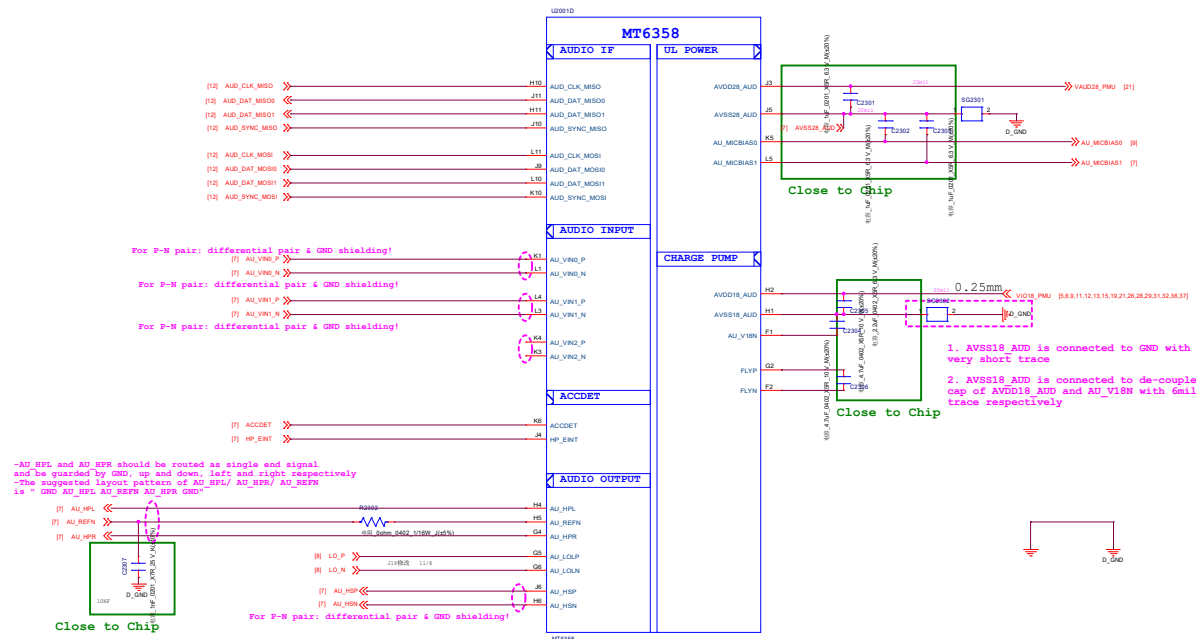
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 19_SENSORS_ALS_PS_A			
Size D	Project PA5905_MB_V1		Rev V1
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Title 09_LCD_CTP_BL		
Size E	Project PA5905_MB_V1	Rev V1
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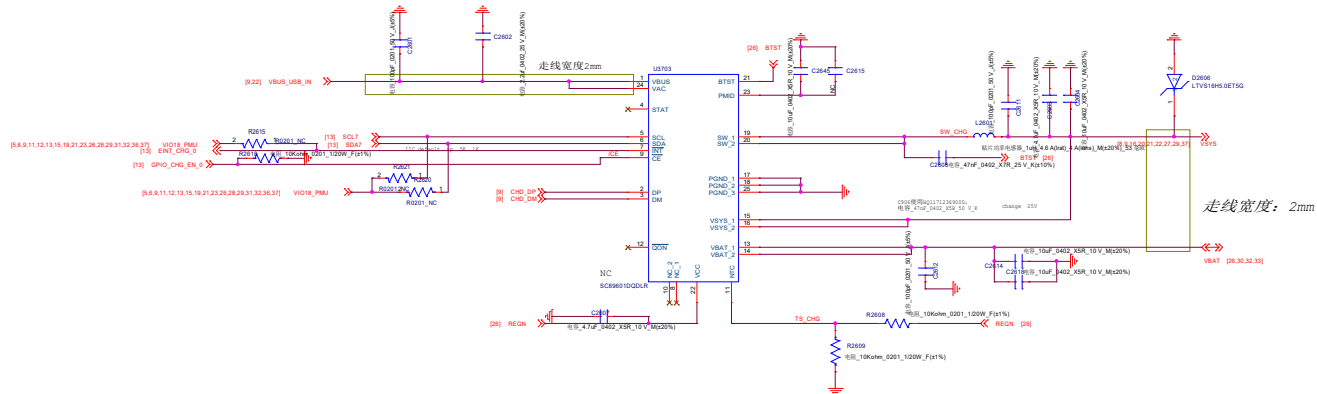


华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 09_LCD_CTP_BL		
Size D	Project PA5905_MB_V1	Rev V1
Date: Wednesday, July 27, 2022	Sheet 22 of 37	

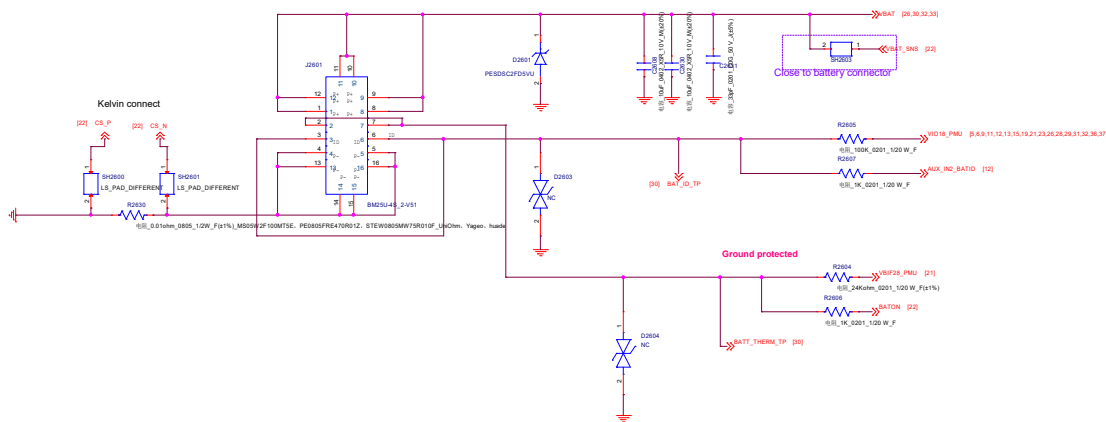


华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 09_LCD_CTP_BL		
Size D	Project PA5905_MB_V1	Rev V1
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SGM41513D Charger



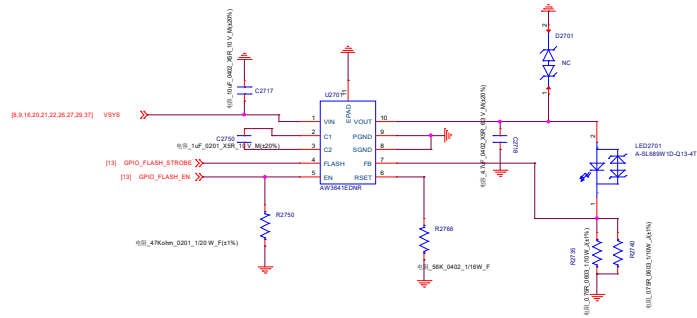
BATTERY CONNECTOR



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 26_POWER_SubPMIC-Charger+CONN			
Size D	Project PA5905_MB_V1	Rev V1	
Date: Friday, September 02, 2022	Sheet 26 of 37		

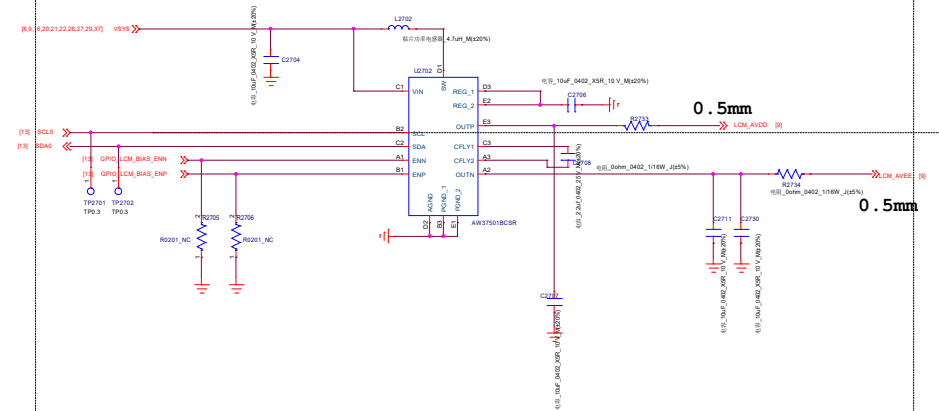
Rear Camera Flash

Flash LED for Camera



Display BIAS

SM5109C I2C Address : 0x3E Write:0x7C / Read:0x7D



华勤通讯 Huaqin Telecom Technology Com.,Ltd

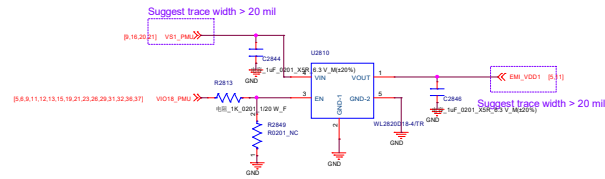
Title 27_CAMERA_INTERFACE

Size D	Project PA5905_MB_V1	Rev v1
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LPDDR4X VDD1 1.8V LDO

Note: 28-5



Schematic design notice of "28_POWER_ThirdParty-Power"

Note 28-1: VA12 Layout placement please close to AP

Note 28-2: VS2 Buck Layout placement please close to PMIC MT6357

Note 28-3: VCN33 LDO Layout placement please close to MT6631

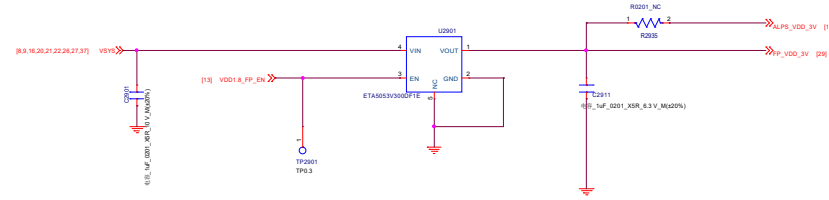
华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 28_POWER_ThirdParty_Powers

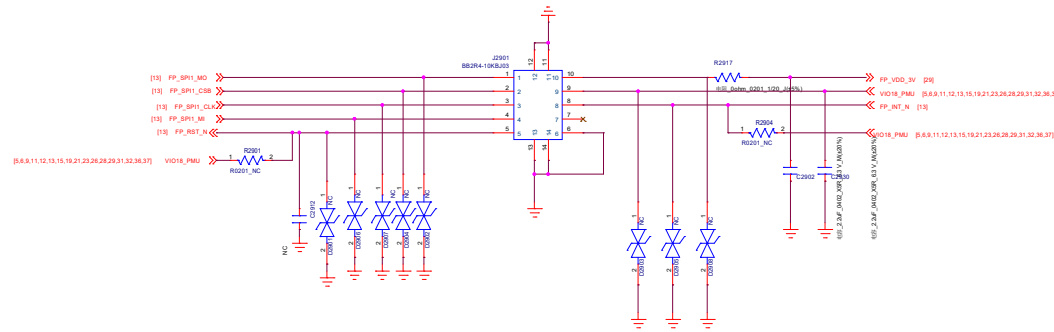
Size D	Project PA5905_MB_V1	Rev v1
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3V LDO Reserve



Fingerprint



序号	引脚 PIN Defina
1	MOST
2	AVDD3.0V (NC)
3	CS
4	DVDD1.8V
5	CLK
6	INT
7	MISO
8	OTP (NC)
9	RST
10	GND

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Title 29_Fingerprint

Size D	Project PA5905_MB_V1	Rev V1
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VCHG



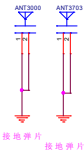
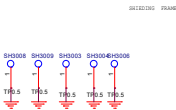
VBAT TEST POINT



UART



RF_PCB TO A-COVER

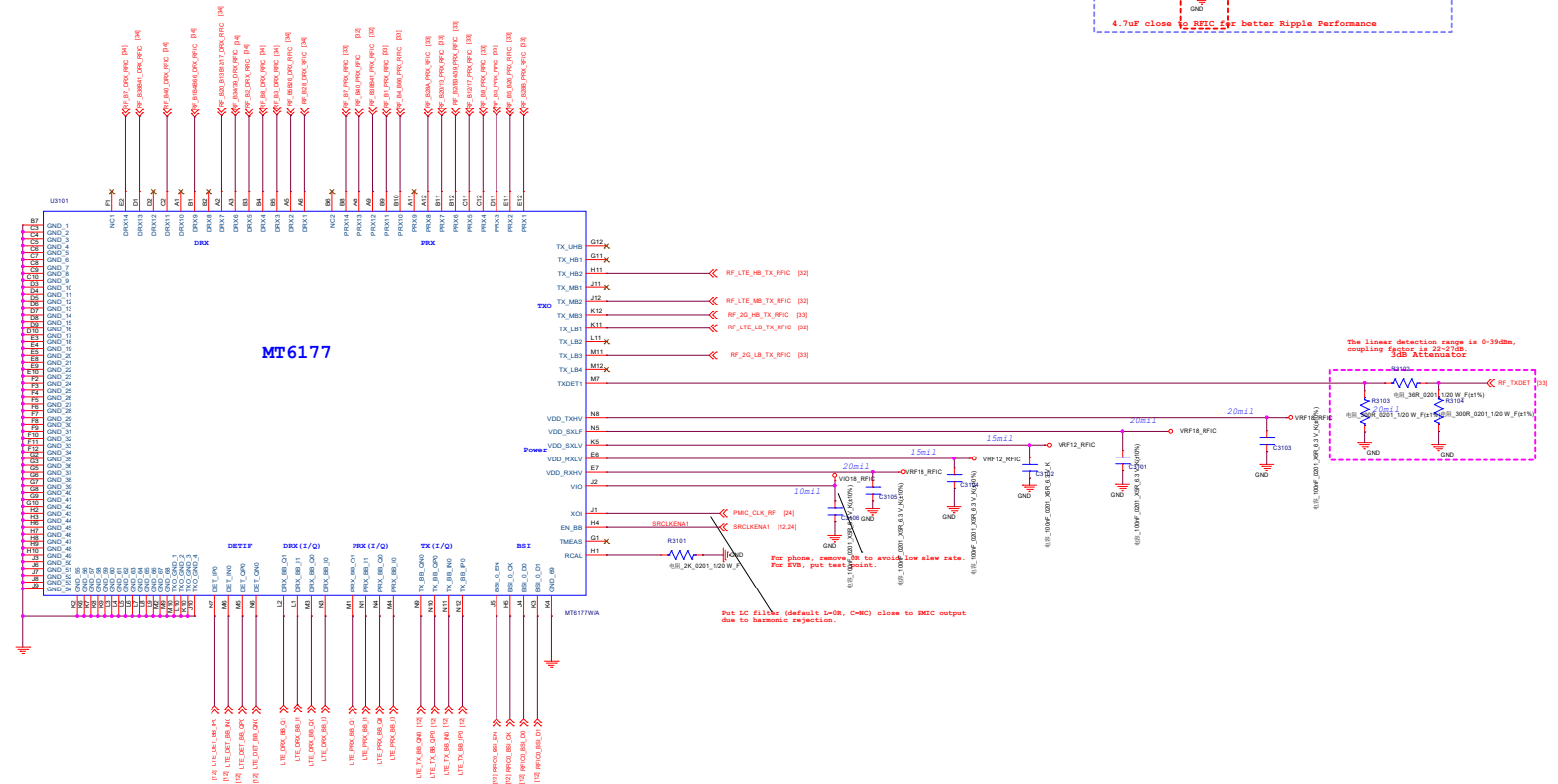


光学定位孔

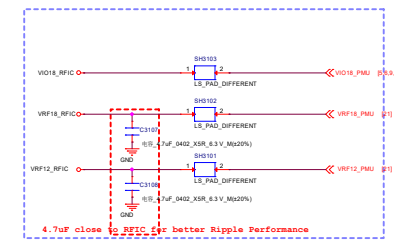


华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 30_DEBUG_IO		
Size D	Project PA5905_MB_V1	Rev V1
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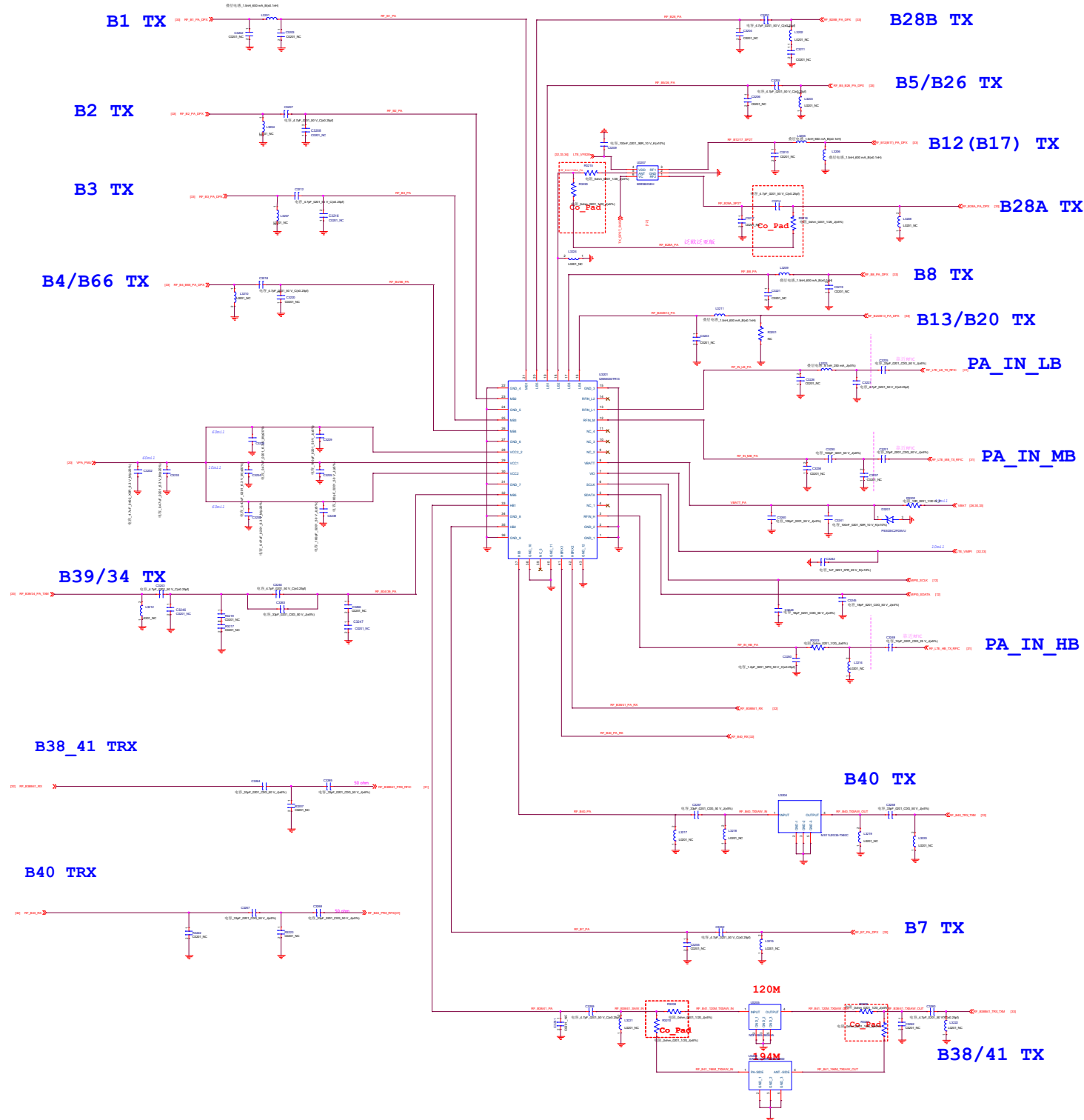
PRX/DRX 1-8 For LB/MB:600M-2025M
PRX/DRX 9-12 For HB/MB:1805M-2690M



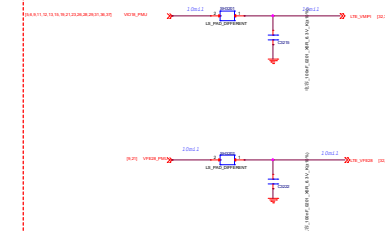
Power domain of MT6177



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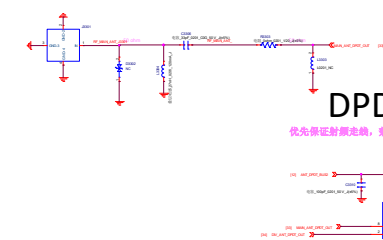
华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 32_RF_MT6177_RF_TX

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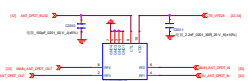
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TXM_GSM

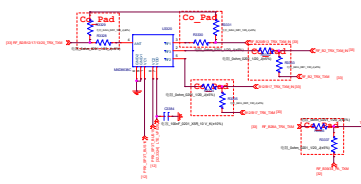


DPDT SWITCH

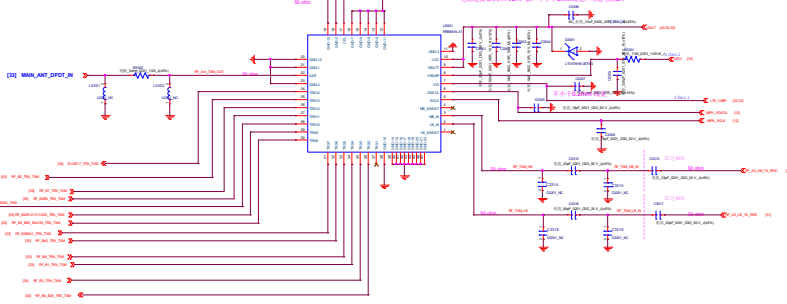
优先保证射频走线, 接零通路, 不要有断头线



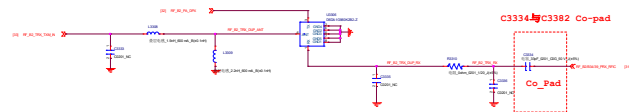
TRX SP3T



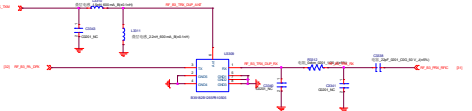
ASM SWITCH



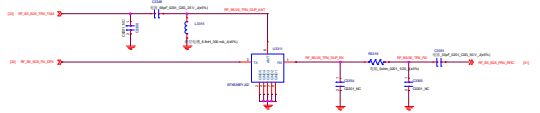
B2/PCS (拉美/国际POCO) DPX



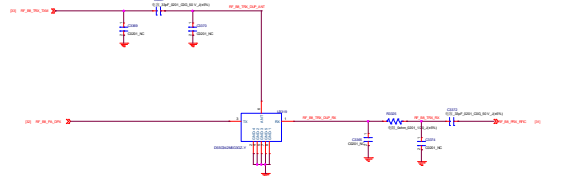
B3 DPX



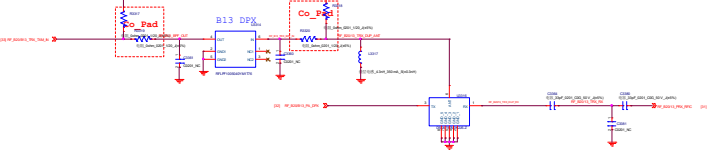
B5/B26 DPX



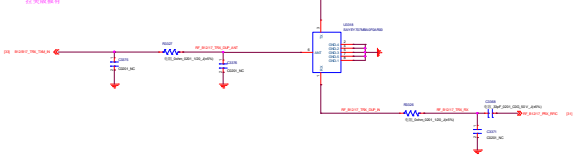
B8 DPX



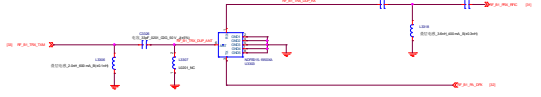
B13/20 DPX



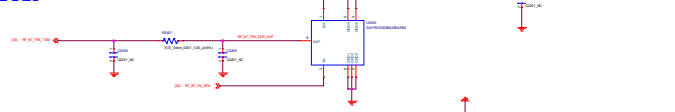
B12/B17 PRX



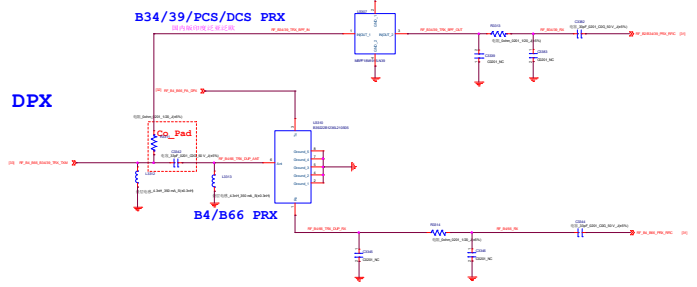
B1 DPX



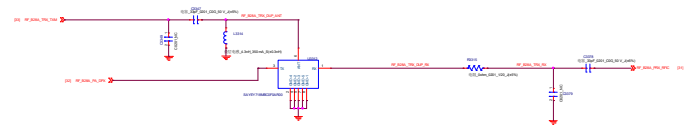
B7 PRX



B4/B66 B34/39/PCS DPX



B28A DPX

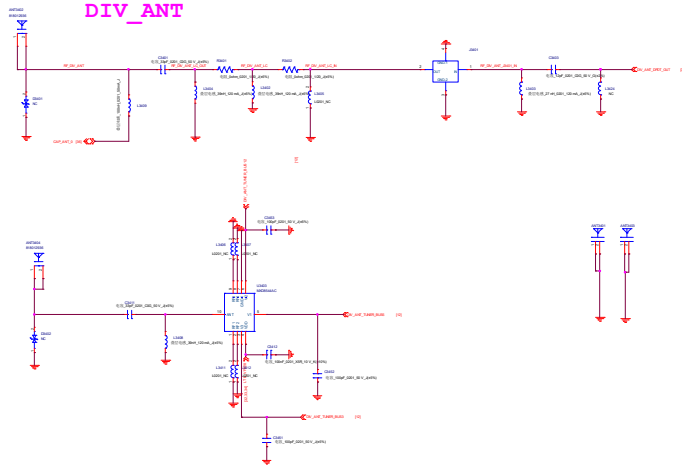


B28B DPX

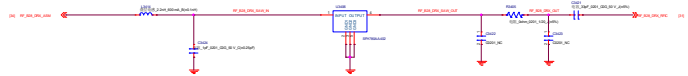


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DIV_ANT

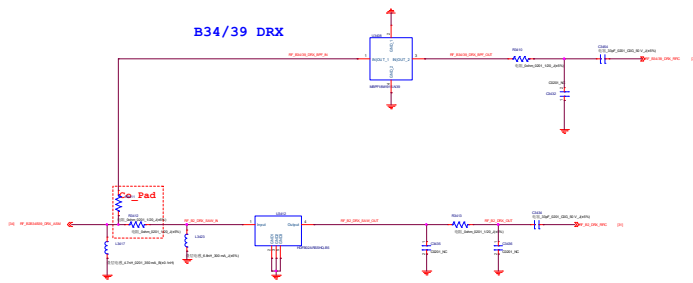


B28 DRX

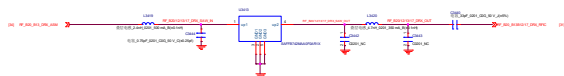


B2 (拉美)/B34/B39 DRX

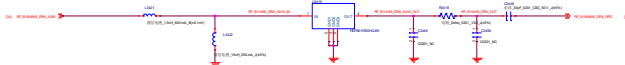
B34/39 DRX



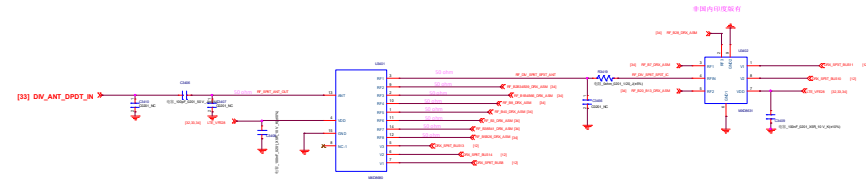
B20/B12 (B17)/B13 DRX



B1/B4/B66 DRX



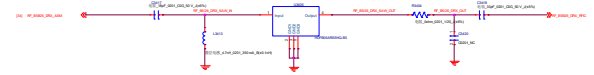
DRX Switch



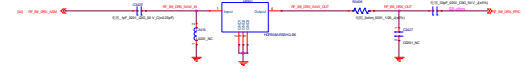
B38/B41 DRX



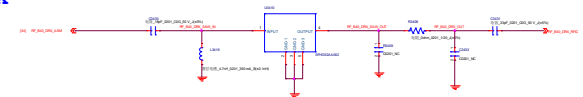
B5/B26 DRX



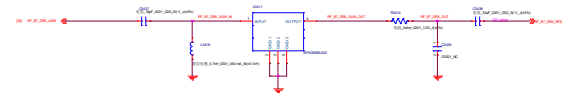
B8 DRX



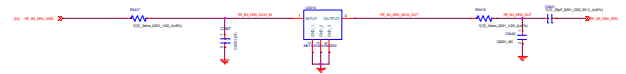
B40 DRX



B7 DRX



B3 DRX



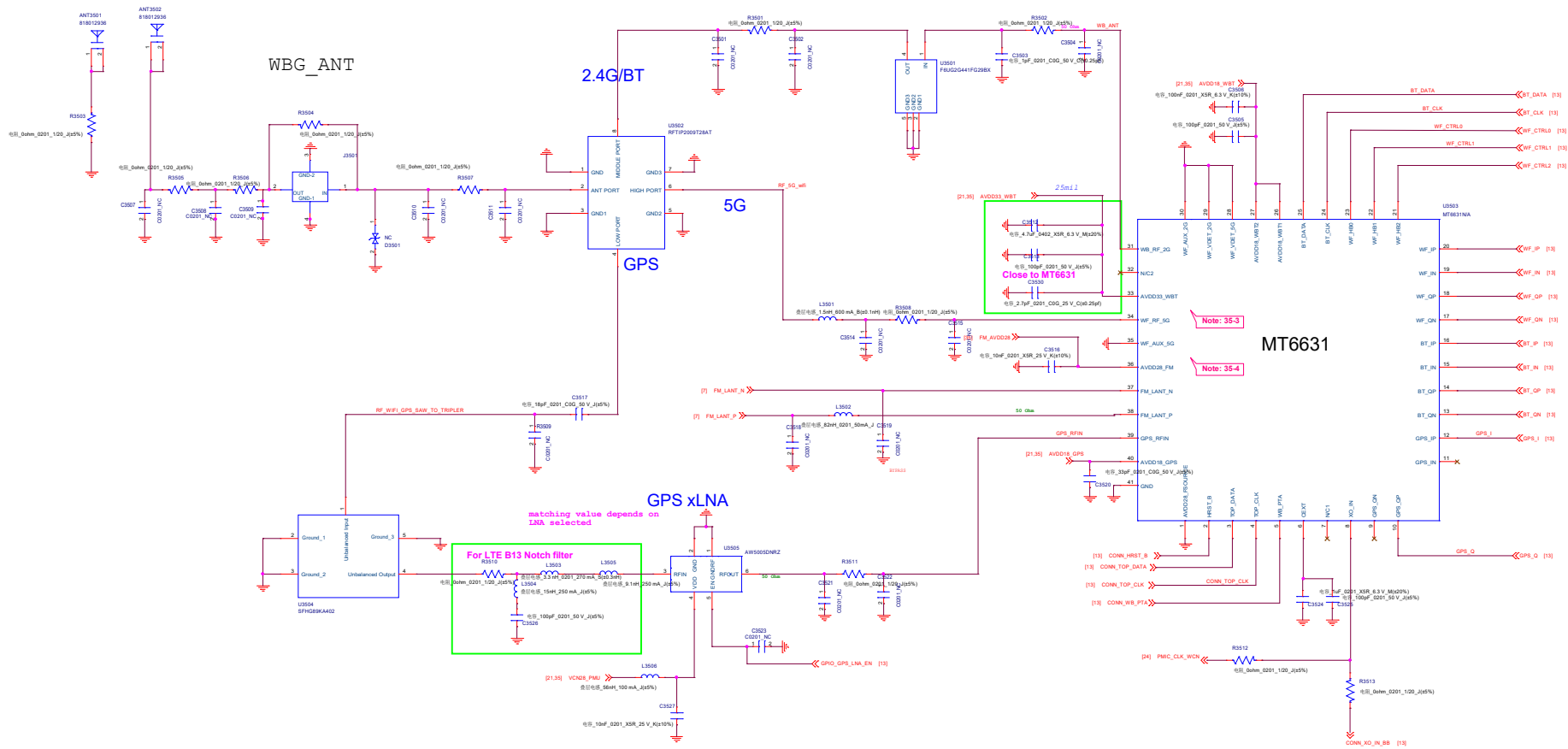
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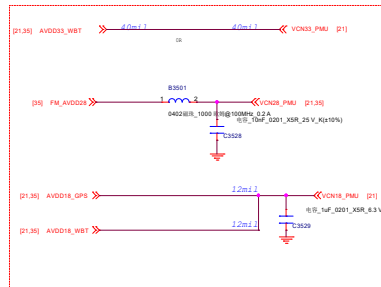
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WCN Power Supply



Schematic design notice of "51_CONNECTIVITY_CONSYS_MT6631"

Note 35-1: For R5015 size, please select 0402 size or larger one

Note 35-2: Please refer to MT6765 Baseband design notice for VCN33 LDO selection guide

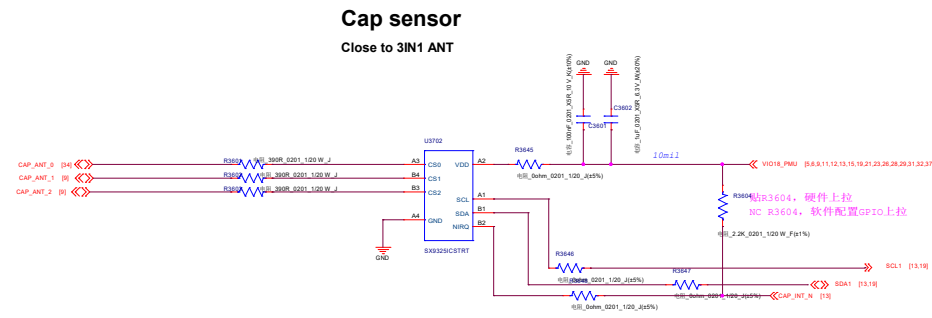
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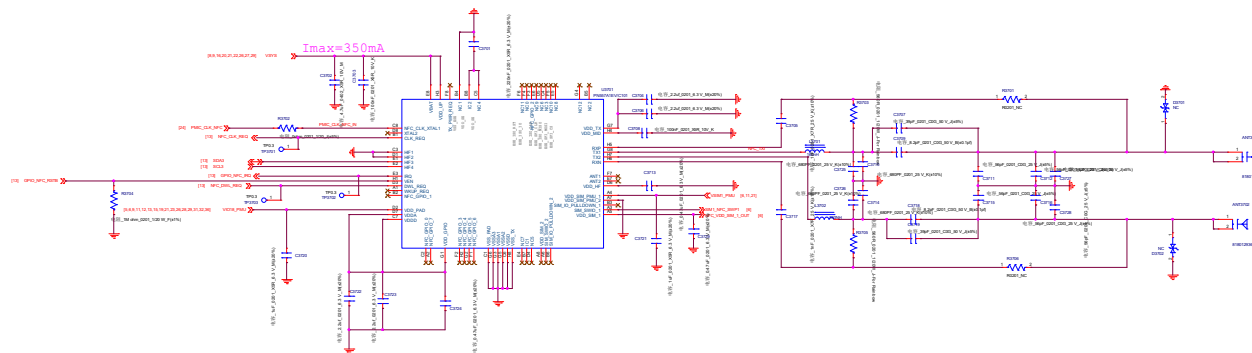
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RF_NFC



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